

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 5: System Software

Subtopic 5.1: Operating Systems

Concept 5.1.4: Program Libraries and DLLs

Total Questions: 11

Mark Schemes begin on Page 7

Question 1 | Source: 9618_w25_qp_11 (Q5.b.i)

- 5 The Operating System (OS) is responsible for hardware and security management in a computer system.

(b) (i) Describe **one** benefit of using library routines when developing software.

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Question 2 | Source: 9618_w25_qp_11 (Q5.b.ii)

- 5 The Operating System (OS) is responsible for hardware and security management in a computer system.

(ii) Describe **one** drawback of using library routines when developing software.

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Question 4 | Source: 9618_w24_qp_11 (Q4.c)

- 4 A student uses a laptop to write a program that is saved as a text file.

(c) The student used a program library when writing their program.

Explain the benefits to the student of using library files when writing a program.

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Question 5 | Source: 9618_w24_qp_12 (Q4.b)

- 4 A program is written in a high-level language by a team of three programmers using an Integrated Development Environment (IDE).

- (b) The programmers created a new program library whilst developing the program.

Describe the benefits to the programmers of creating a program library.

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Question 6 | Source: 9618_s24_qp_12 (Q8.c)

- 8 A programmer uses an Integrated Development Environment (IDE) to write a computer program. The IDE has both a compiler and an interpreter as built-in translators.

- (c) The programmer uses program libraries when developing the program.

Describe **two** benefits to the programmer of using program libraries.

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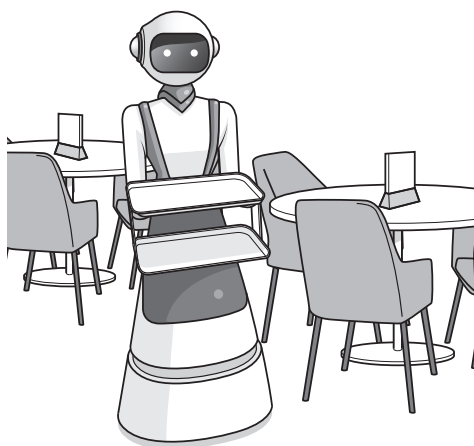
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Question 7 | Source: 9618_s24_qp_13 (Q7.e.i)

- 7 Robots are used to serve food and drink to customers at a restaurant.



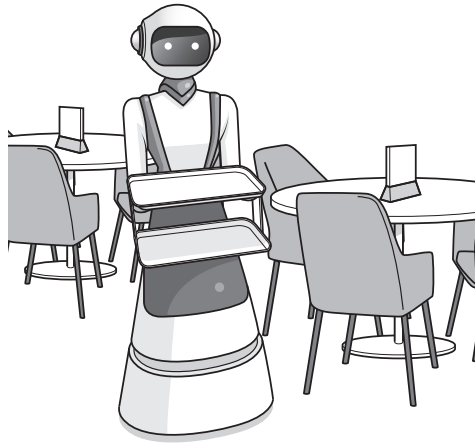
- (e) Program libraries were used when writing the robot's software.

- (i) State what is meant by a **program library**.

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Question 8 | Source: 9618_s24_qp_13 (Q7.e.ii)

- 7 Robots are used to serve food and drink to customers at a restaurant.



- (e) Program libraries were used when writing the robot's software.

- (ii) Some program libraries include Dynamic Link Library (DLL) files.

Describe the benefits of a programmer using a library with DLL files instead of using a library that does not include DLL files.

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Question 9 | Source: 9618_s23_qp_12 (Q7.a.i)

- 7 A software developer is working in a team writing a program for a client.

- (a) The developer is writing a new program library to be used by the other team members.

- (i) Define the term **program library**.

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Question 10 | Source: 9618_s23_qp_12 (Q7.a.ii)

- 7 A software developer is working in a team writing a program for a client.

- (a) The developer is writing a new program library to be used by the other team members.

- (ii) Explain **two** benefits to the developer of choosing to create a Dynamic Link Library (DLL).

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Question 11 | Source: 9618_w22_qp_11 (Q7.a)

- 7 (a) State **two** benefits to a programmer of using Dynamic Link Library (DLL) files.

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Question 12 | Source: 9618_s21_qp_12 (Q7.a)

- 7 Jennifer is writing a computer program for her A Level homework.

- (a) Jennifer uses a program library to help her write her computer program.

Describe how a program library can be used while writing a computer program.

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MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q5.b.i)

5(b)(i)	One mark for benefit, one mark for appropriate matching expansion e.g. • Less code needs to be developed as code is pre-written • ... thereby reducing the time to write a program • The library routine will be pre-tested and documented • ... reducing the time taken to test a program // giving confidence that the routine works as it should • The code for the library routine can be written in a different programming language • ... which allows the use of special features of that language • The user's source code is more readable • ... as the library routine is simply called in the source code rather than being written out in full • The library could contain the code for complex routines • ... thus allowing the programmer to include modules they might not be able to code	2
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Mark Scheme for Question 2 | Source: 9618_w25_ms_11 (Q5.b.ii)

5(b)(ii)	1 mark for drawback, 1 mark for appropriate matching expansion e.g. • Compatibility issues • ... the library routine may not work with current program • Not guaranteed to be thoroughly tested • ... there could be unexpected problems (bugs/virus) • The library routine may not match needs exactly // The library routine may give unexpected results • ... so it will need to be edited and tested • If the library routine is later edited or changed (by someone else) • ... there could be unexpected errors or results	2
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Mark Scheme for Question 3 | Source: 9618_w25_ms_13 (Q8.b)

8(b)	1 mark per bullet point, max 3 marks • Programming time is saved as code does not have to be written from scratch • Testing time is saved as code is already tested / documented • A library routine is more likely to work, as code is already tested • Library routines automatically update if they are changed / improved • The programmer can use library routines to perform complex functions / procedures that they may not be able to write themselves	3
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Mark Scheme for Question 4 | Source: 9618_w24_ms_11 (Q4.c)

4(c)	1 mark for each bullet point (max 3) e.g. • It saves development time for the student • ... by using pre-written code • There is less testing to do // the routines are more likely to work correctly • ... because they have already been tested and are error-free • It results in a more robust program • Subroutines can be used that they do not have the skill to write • ... which enables a more complex program to be written	3
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Mark Scheme for Question 5 | Source: 9618_w24_ms_12 (Q4.b)

4(b)	1 mark for each bullet point (max 3) e.g. • Subroutines can be shared / reused • ... between team members who are working independently • ... without having to rewrite/re-test them which saves the programmers' time • A program library provides continuity between programs/programmers • Individual programmers can contribute their specialisms to the library // Individual programmers can use the specialisms of others	3
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Mark Scheme for Question 6 | Source: 9618_s24_ms_12 (Q8.c)

8(c)	1 mark each to max 2 : • Saves programming/testing time as code does not have to be written/re-written from scratch // code does not have to be tested • Code is already tested so it is more robust/likely to work • The programmer does not need to maintain the library // library routines are updated automatically • Can perform complex calculations that the programmer may be unable to do • Makes code more easily readable	2
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Mark Scheme for Question 7 | Source: 9618_s24_ms_13 (Q7.e.i)

7(e)(i)	1 mark for: Pre-written code/functions/routines that can be imported/called in another program	1
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Mark Scheme for Question 8 | Source: 9618_s24_ms_13 (Q7.e.ii)

7(e)(ii)	1 mark each to max 4 : - Maintenance not needed to be done by the programmer ... because the DLL is separate from program - The calling program does not need recompilation by the programmer when a DLL file changes ... because the DLL file can be updated independently of the calling program ... updates will apply to all programs that use the DLL file	4
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Mark Scheme for Question 9 | Source: 9618_s23_ms_12 (Q7.a.i)

7(a)(i)	1 mark each to max 2 - Set of pre-written / pre-compiled / pre-tested subroutines ... which can be called in other programs ... by installing/importing the library	2
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Mark Scheme for Question 10 | Source: 9618_s23_ms_12 (Q7.a.ii)

7(a)(ii)	1 mark for each bullet point. Mark in pairs; 1 mark for a benefit and 1 mark for an appropriate expansion (main) memory requirements for program is reduced ... as dynamic link library is loaded only once / when required - the executable file size of the program using the DLL will be smaller ... because the executable does not contain (all) the library routines - maintenance not needed to be done by the programmer ... because the DLL is separate from program - no need to recompile the main program when changes are made to DLL ... because changes / improvements/ error correction to the DLL file code are done independently of the main program - A single DLL file can be made available to several application programs ... Saving space in memory / easing the pressure on memory	4
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Mark Scheme for Question 11 | Source: 9618_w22_ms_11 (Q7.a)

7(a)	1 mark for each benefit (max 2): • (main) memory requirements for program are reduced as dynamic link library is loaded only once / when required • the executable file size is smaller because the executable does not contain all the library routines • maintenance not needed to be done by the programmer because the DLL is separate from program • no need to recompile the main program when changes are made to DLL because changes / improvements/ error correction to the DLL file code are done independently of the main program	2
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Mark Scheme for Question 12 | Source: 9618_s21_ms_12 (Q7.a)

7(a)	1 mark per bullet point to max 2 • Program libraries store pre-written functions and routines • The program library can be referenced/imported • the functions/routines can be called in her own program	2
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